

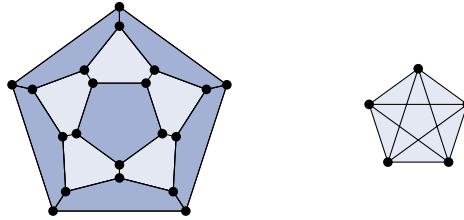
that transforms one copy of an edge to another passes over a vertex. Interestingly, the upper bound for 3-planar graphs is tight in this more general setting only [4, 6].

The *crossing number* of a drawing  $\Gamma$  is the number of edge crossings in  $\Gamma$ . The *crossing number*  $\text{cr}(G)$  of a graph  $G$  is the minimum crossing number over all drawings of  $G$ . By definition every  $k$ -planar graph  $G$  admits a  $k$ -plane drawing and thus

$$\text{cr}(G) \leq \frac{km}{2}, \quad (\text{S})$$

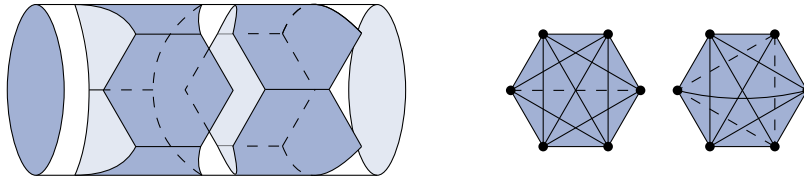
where  $m$  denotes the number of edges in  $G$ . For a  $k$ -planar graph, this simple inequality connects upper bounds on the number of edges with lower bounds on the crossing number. Both of these come together in the well-known Crossing Lemma [2, Chapter 45], as the best constants in the Crossing Lemma are obtained by analyzing  $k$ -plane drawings [1, 6, 10, 11]. Conversely, combining the lower bound on  $\text{cr}(G)$  from the Crossing Lemma with an upper bound on  $\text{cr}(G)$  we obtain an upper bound on the number of edges in  $G$ . While (S) would work here, it is probably not an ideal choice because the graphs for which (S) is tight might be very different from those graphs that have a maximum number of edges, for any fixed  $n$ . For instance, for a 1-planar graph  $G$  we have  $\text{cr}(G) \leq n - 2$  [17, Proposition 4.4], which beats the bound we get by plugging  $m \leq 4n - 8$  into (S) by a factor of two. Can we obtain similar improvements by bounding  $\text{cr}(G)$  in terms of  $n$ , rather than  $m$ , for  $k \geq 2$ ?

Indeed, very recently it has been shown that  $\text{cr}(G) \leq 3.3n$  if  $G$  is 2-planar and  $\text{cr}(G) \leq 6.6n$  if  $G$  is 3-planar [3]. There is some indication that the bound for 2-planar graphs could be tight up to an additive constant, as it is achieved by the standard drawings of optimal 2-planar graphs (Figure 1). But the crossing number of these graphs is not known.



■ **Figure 1** Construction by Pach and Tóth [15, Figure 3]. Left: A planar drawing with pentagonal faces. Right: To each pentagonal face all diagonals are added.

In contrast, there exists a family of simple 3-planar graphs with  $5.5n - 15$  edges whose standard drawings have  $5.5n - 21$  crossings (Figure 2). Thus, there is a gap of  $1.1n$  between the lower and the upper bound for the crossing number of 3-plane drawings.



■ **Figure 2** Construction from [11, Figure 8]. Left: A cylinder with two layers, each consisting of three hexagonal faces. Right: To each face of a layer all but one diagonal is added. To the top and bottom face six diagonals are added. Missing diagonals are represented by dashed lines.